GROSS ANATOMY AND HISTOLOGY OF LUNG

LEARNING OBJECTIVES

At the end of lecture the student should be able to :

- **The detailed structure of right and left lungs and the difference between them**
- **The root of lung and the structures forming it**
- **The bronco pulmonary segments and their importance**
- **Pulmonary and bronchial vessels**
- **The lymphatic system of each lung**
- **The nerve supply to lungs , pulmonary plexus and the importance of phrenic nerve**
- **Discuss the histological features of extra and intra pulmonary part of bronchial tree**

LUNGS

- Essential organ of respiration
- Conical in shape
- Each lung has

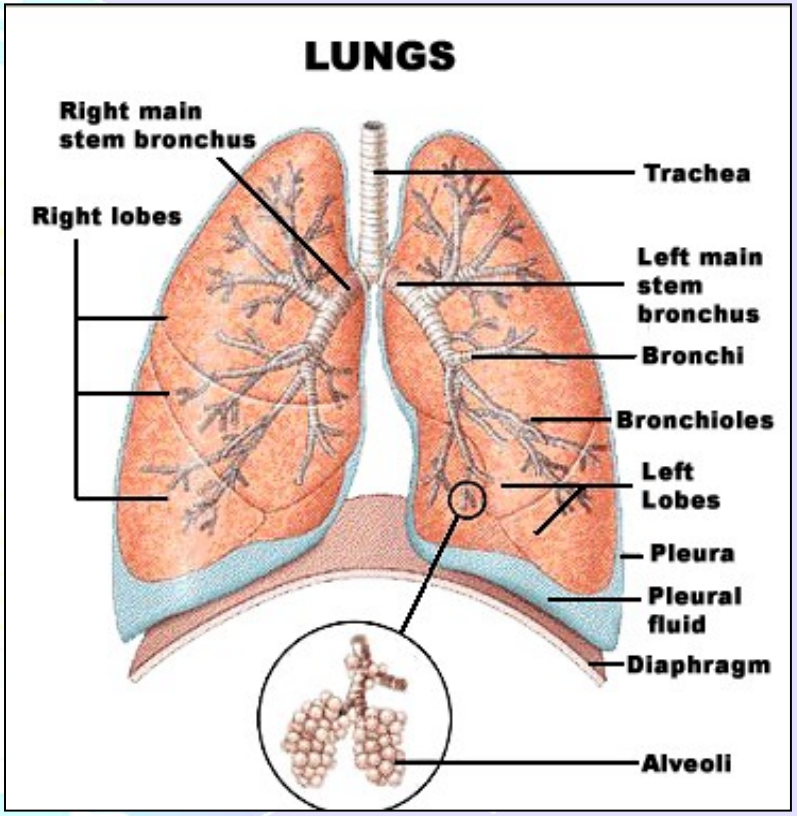
  an apex,

  base,

  3 borders(anterior,posterior inferior)and

  2 surfaces(costal and medial)

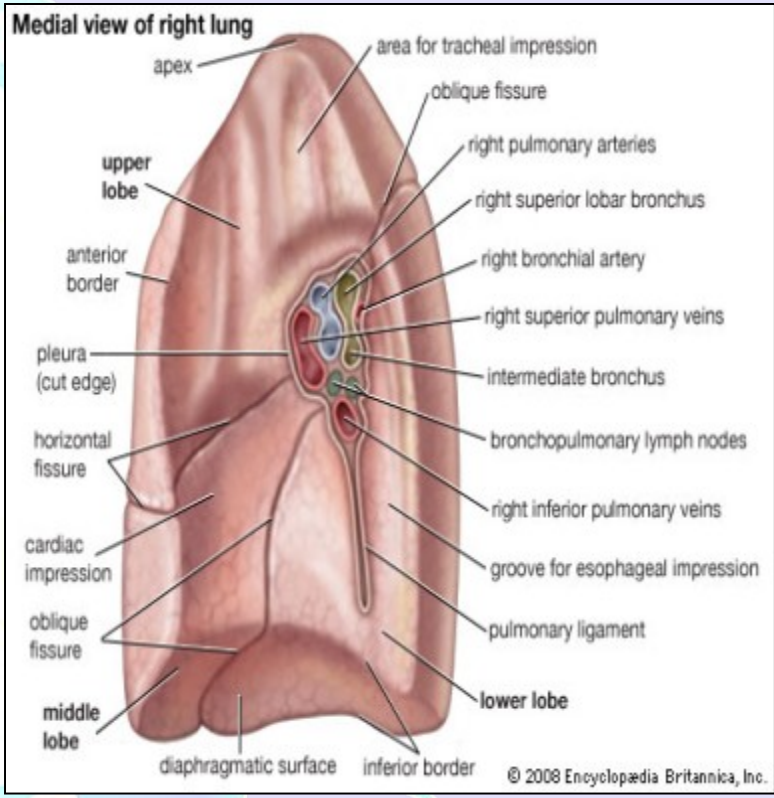
- Medial surface is divided into vertebral and mediastinal parts



LUNGS

Apex:

- Blunt
- Covered by cervical pleura and suprapleural membrane.
- Is grooved by subclavian artery on the medial side and in front.



Base:

- Semilunar and concave
- Rest on diaphragm.

Anterior border:

- Very thin, shorter than posterior border.
- On right side vertical and on left side has a wide cardiac notch .

LUNGS

Posterior border:

Is thick and ill defined

Inferior border:

Separates the base from the costal and medial surfaces

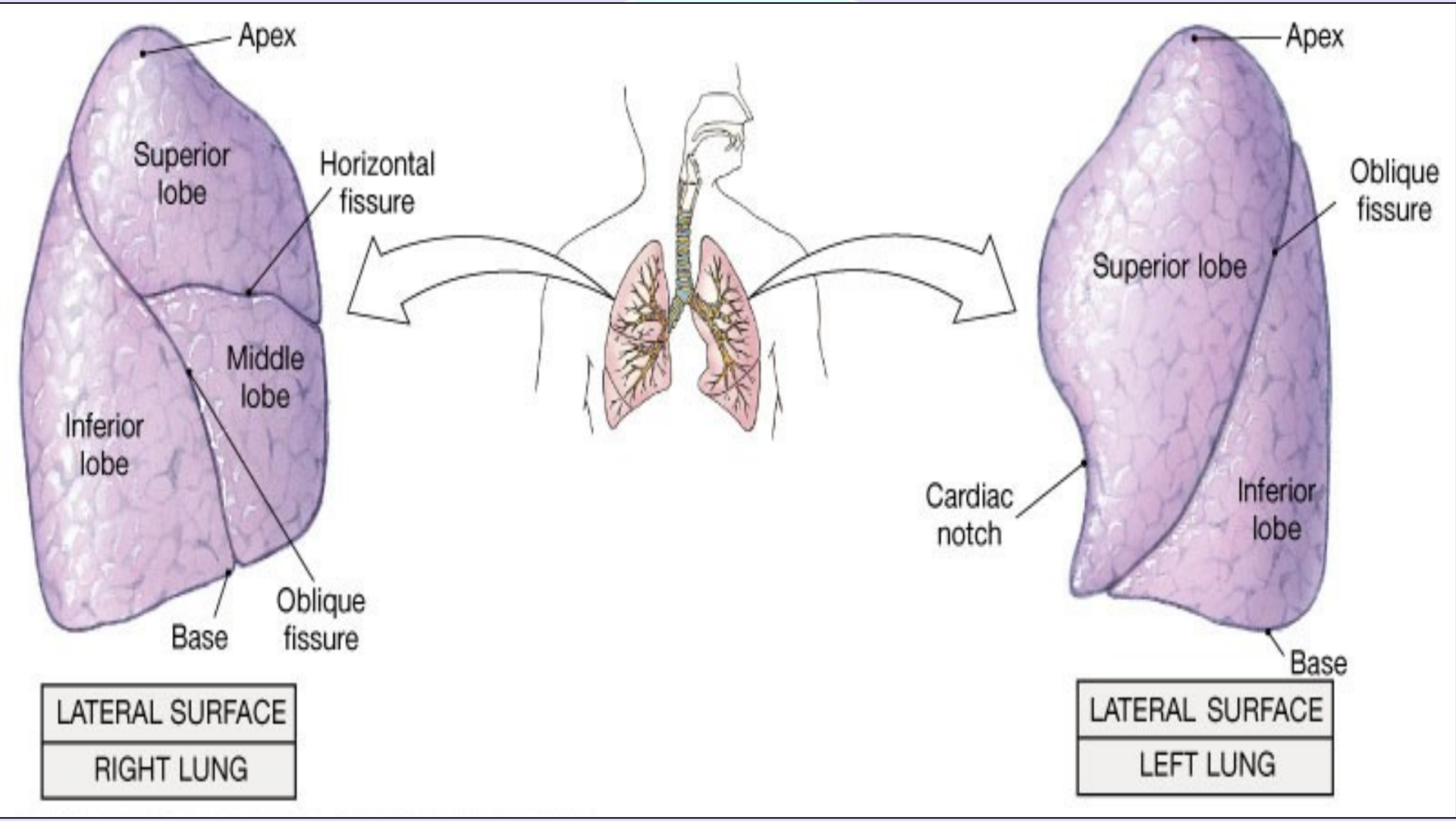
Costal surface:

Is large and convex.

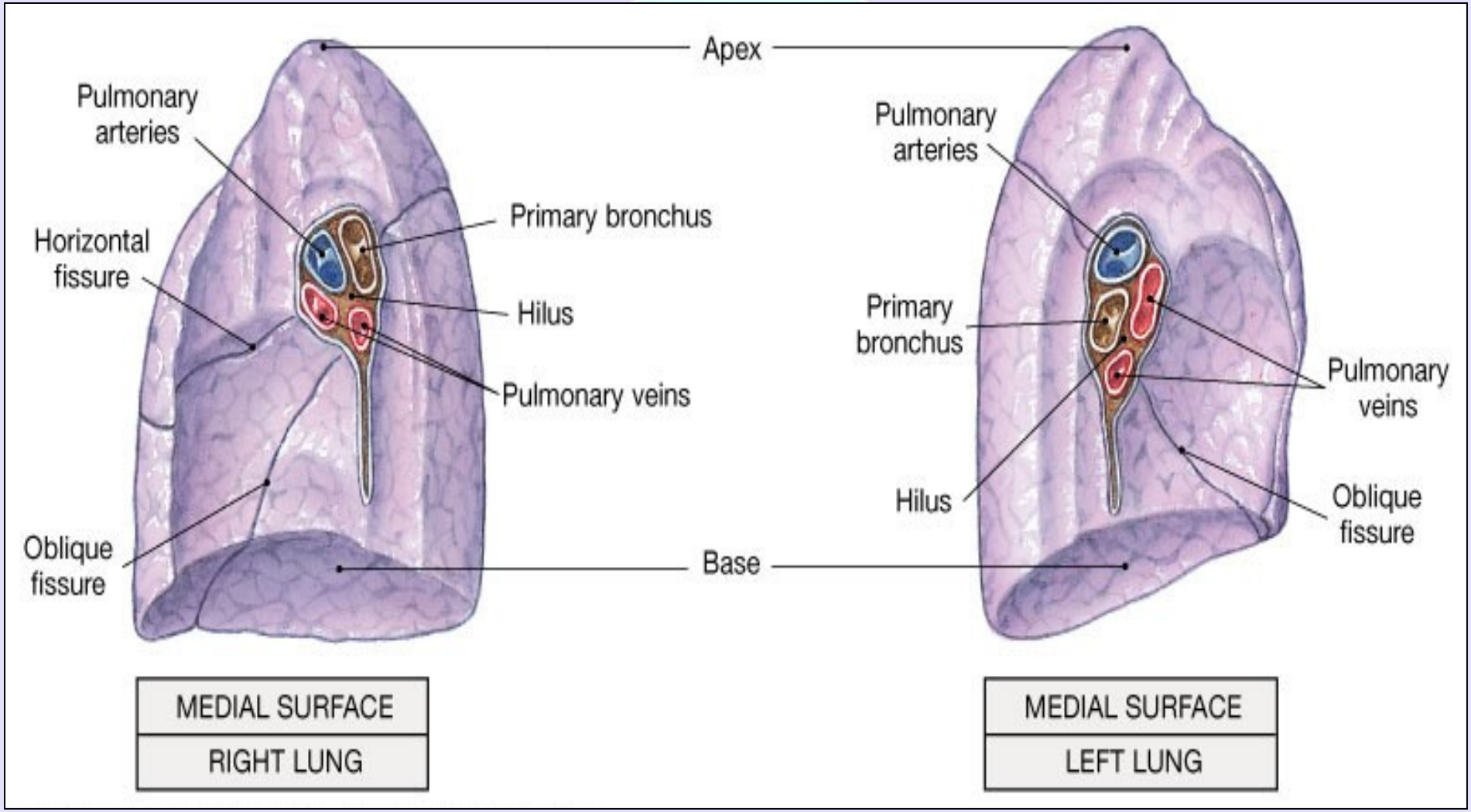
Medial surface:

divided into posterior or vertebral part, and an anterior or mediastinal part.

Gross Anatomy of the Lungs

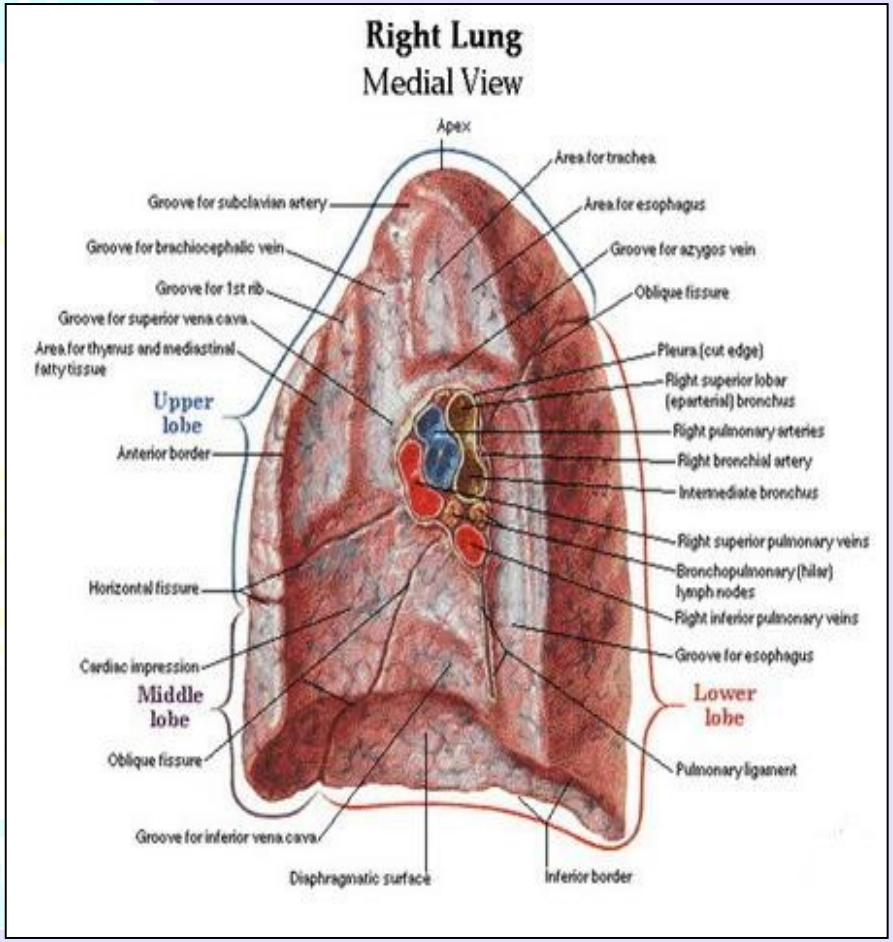


Gross Anatomy of the Lungs



RIGHT LUNG

- **Has an apex that projects into the root of the neck**
- **The apex is smaller than that of the left lung**
- **Is larger and heavier than the left lung, but is shorter and wider because of the higher right dome of diaphragm and the inclination of the heart to the left.**



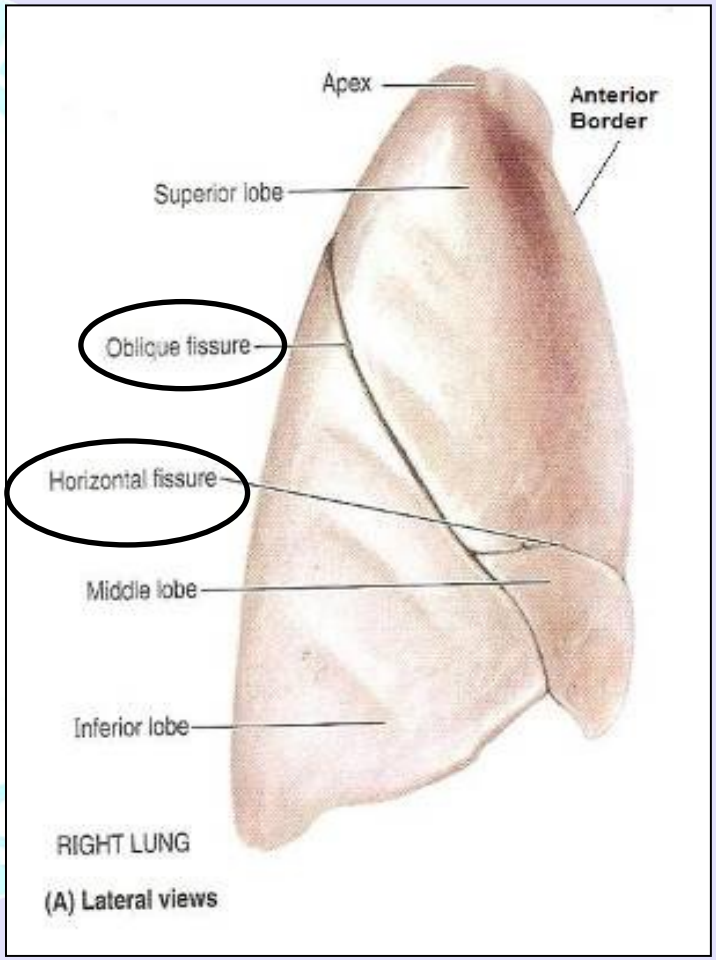
RIGHT LUNG

Is divided into

- upper
- middle and
- lower lobes

By the oblique and horizontal (accessory) fissures but usually receives a single bronchial artery.

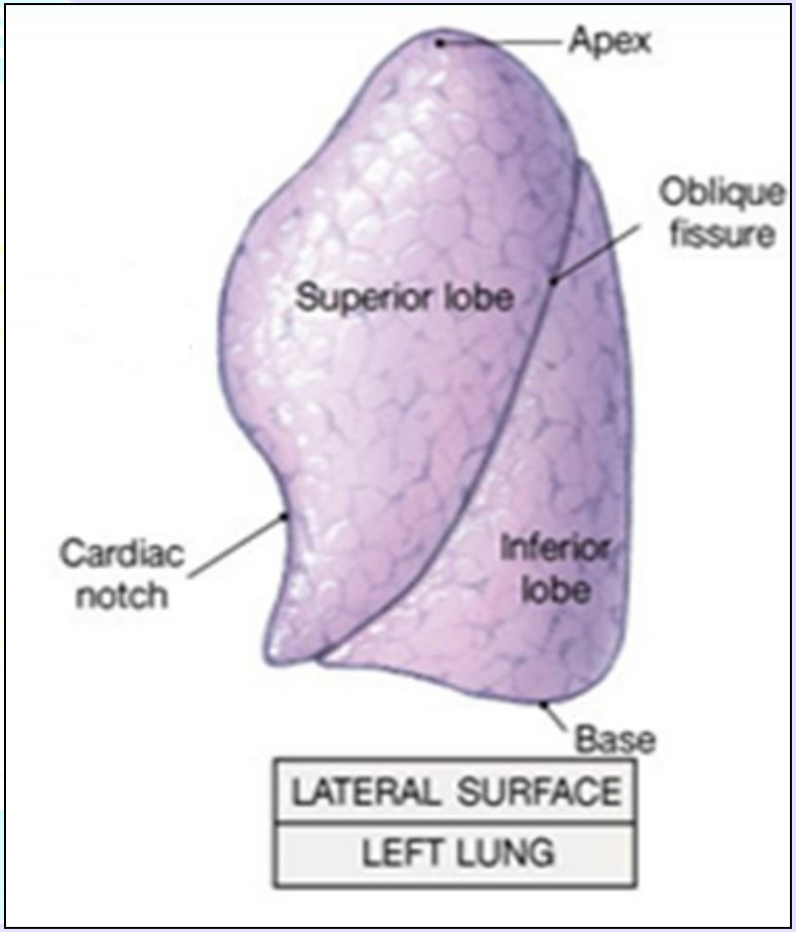
- The oblique fissure usually begins at the head of fifth rib and follows roughly the line of sixth rib



LEFT LUNG

- Is divided into upper and lower lobes

  by an oblique fissure that follows the line of the sixth rib
- Is usually more vertical in the left lung than in the right lung
- Usually receives two bronchial arteries
- Contains lingula, tongue shaped portion of the upper lobe that corresponds to the middle lobe of the right lung

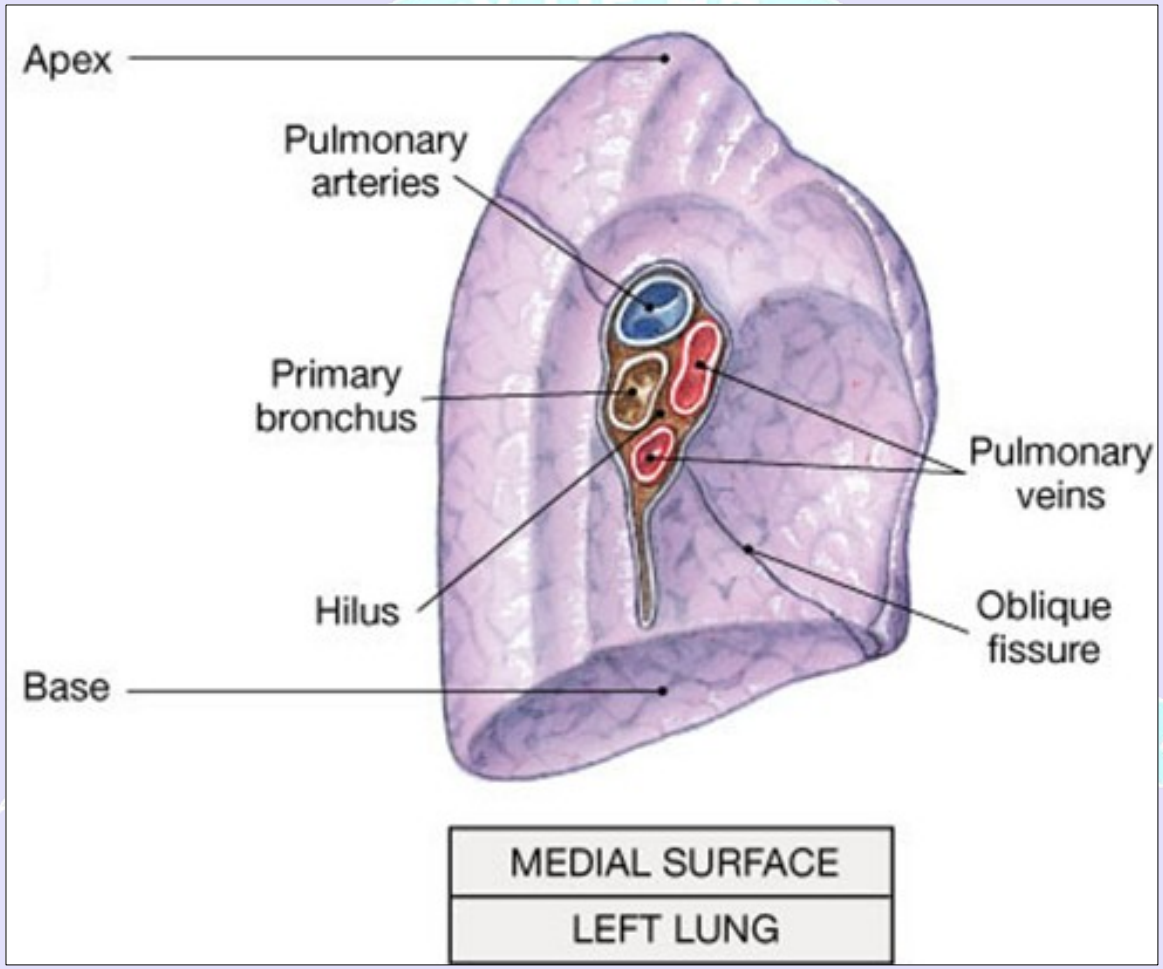


LEFT LUNG

- Contains a cardiac impression ,a cardiac notch, and grooves for aortic arch, descending aorta and left subclavian artery

- Has two lobar(secondary) bronchi and 8-10 segmental bronchi

LEFT LUNG


Apex
Pulmonary arteries
Primary bronchus
Hilus
Base
Pulmonary veins
Oblique fissure
MEDIAL SURFACE
LEFT LUNG

ROOT OF THE LUNG

- It is short, broad pedicle which connects the medial surface of the lung to the mediastinum
- Lies opposite to the bodies of 5 th,6th and 7th thoracic vertebrae
- Formed by structures which either enter or come out of the lung at hilum

ROOT OF THE LUNG

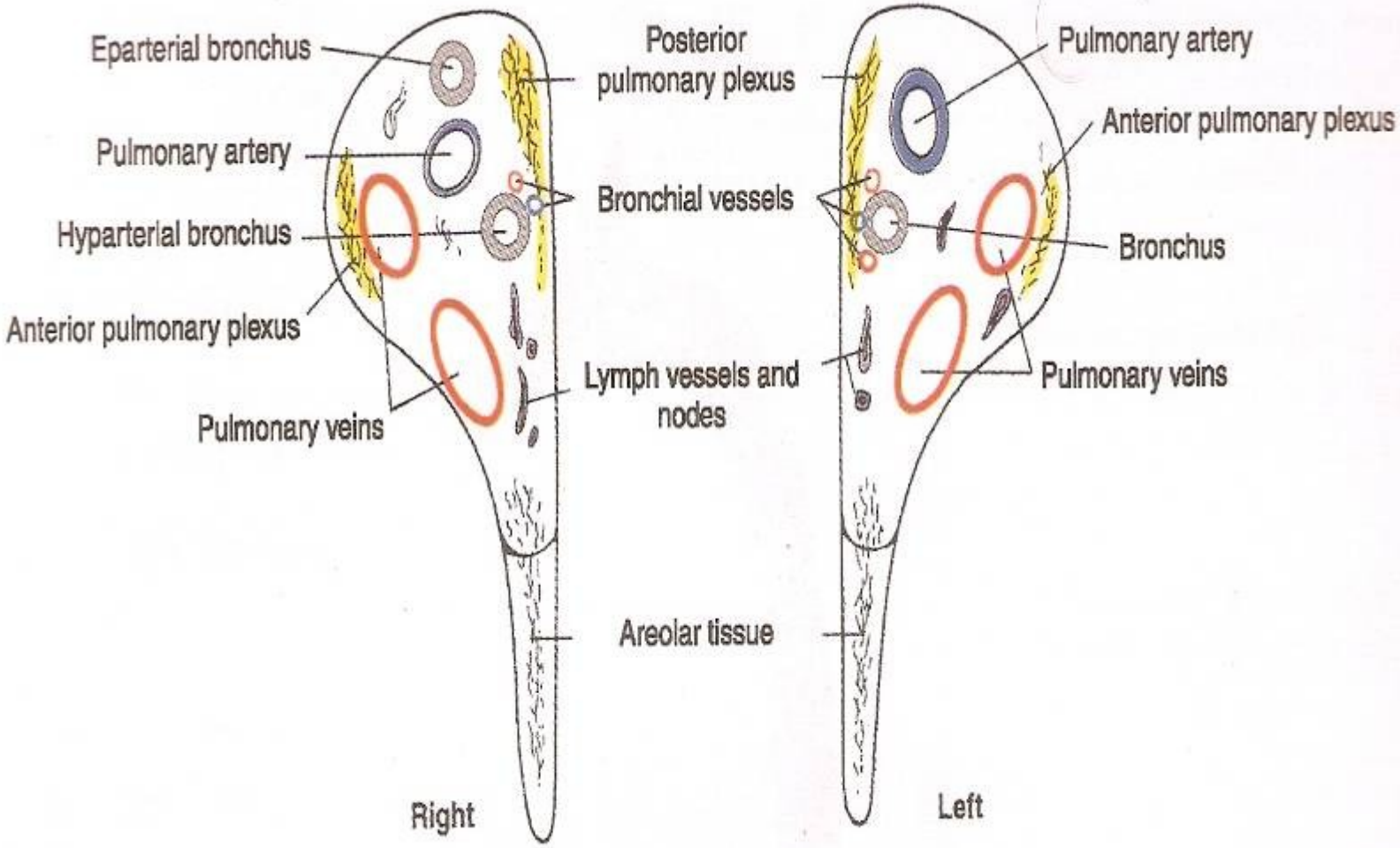


Fig. 16.4: Roots of the right and left lungs seen in section.

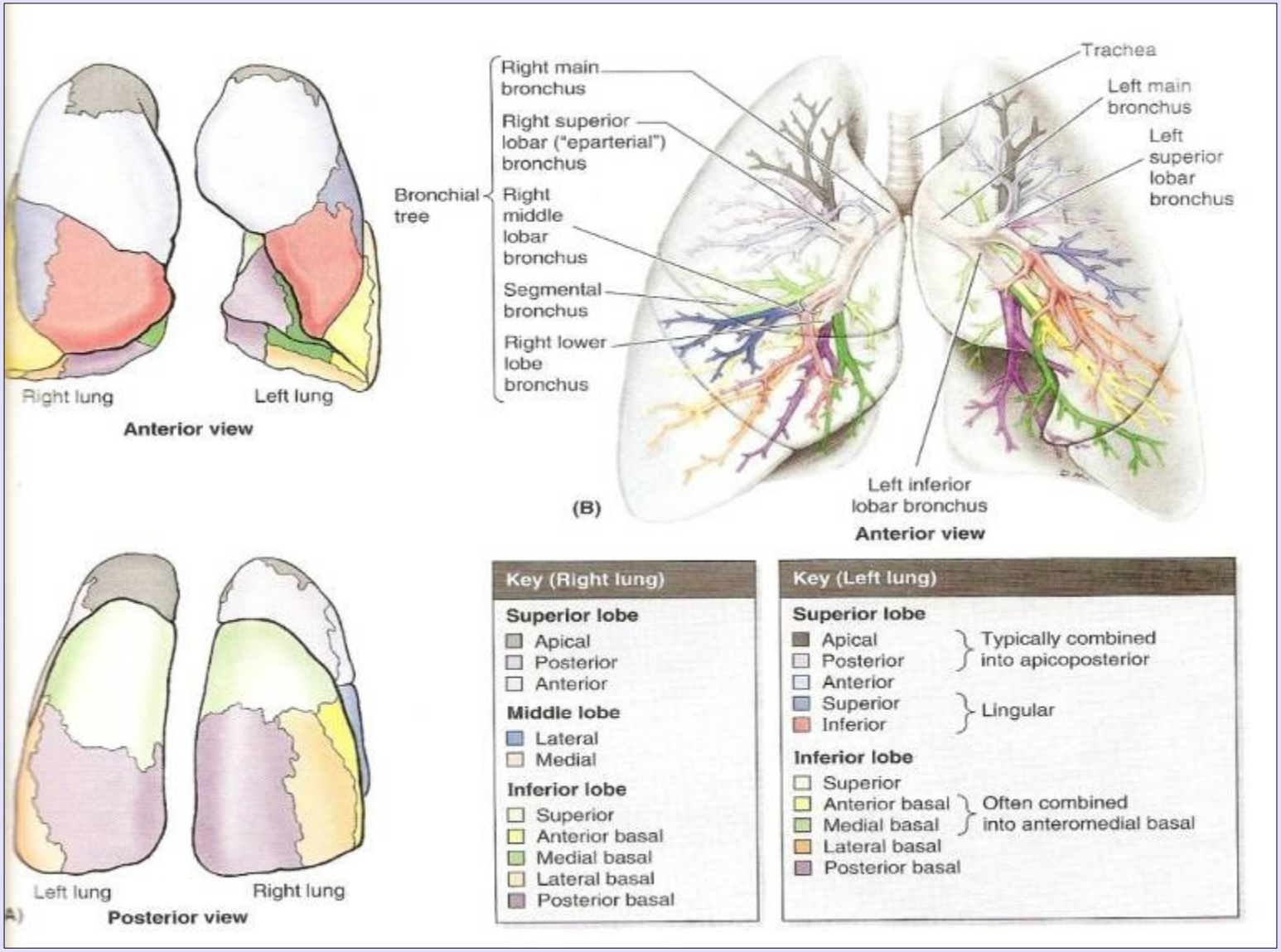
BRONCHOPULMONARY SEGMENT

- Is the anatomic, functional and surgical unit of the lungs
- Consists of segmental bronchus , a segmental branch of the pulmonary artery ,and a segment of the lung tissue, surrounded by a delicate connective tissue septum.
- It is drained by the intersegmental part of the pulmonary vein

BRONCHOPULMONARY SEGMENT

- Refers to the portion of the lung supplied by each segmental bronchus and segmental artery.
- The pulmonary veins are intersegmental
- Is clinically important because the intersegmental pulmonary veins form surgical landmarks
- Therefore a surgeon can remove a bronchopulmonary segment without seriously disrupting the surrounding lung tissue and major blood vessels

BRONCHOPULMONARY SEGMENT

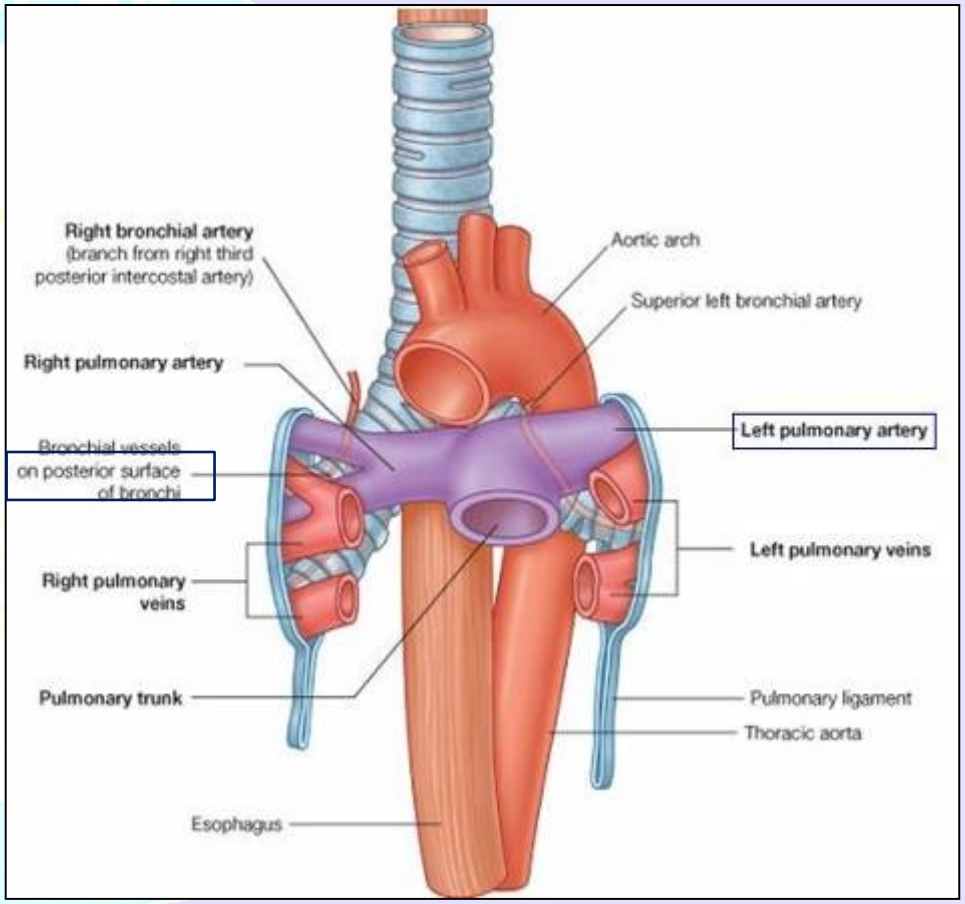


BLOOD VESSELS

PULMONARY TRUNK

- Extends upwards from conus arteriosus of the right ventricle of the heart and carries poorly oxygenated blood to the lungs for oxygenation

- Bifurcates into right and left pulmonary arteries at the level of the sternal angle



BLOOD VESSELS

LEFT PULMONARY ARTERY:

- Carries deoxygenated blood to the left lung
- Is shorter and narrower than the right pulmonary artery
- It arches over the left pulmonary bronchus
- Is connected to the arch of aorta by ligamentum arteriosum, the fibrous remains of the ductus arteriosus

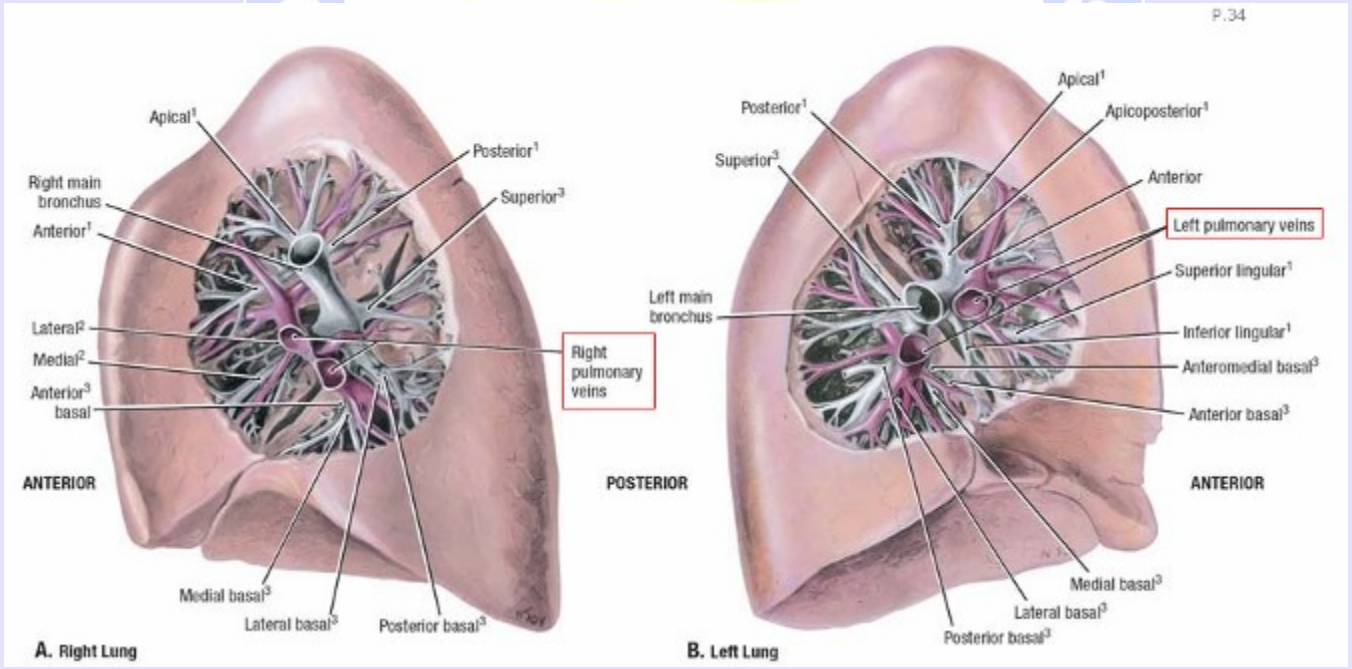
BLOOD VESSELS

RIGHT PULMONARY ARTERY:

- Runs horizontally towards the hilum of the right lung under the arch of aorta behind the ascending aorta and SVC and anterior to the right bronchus

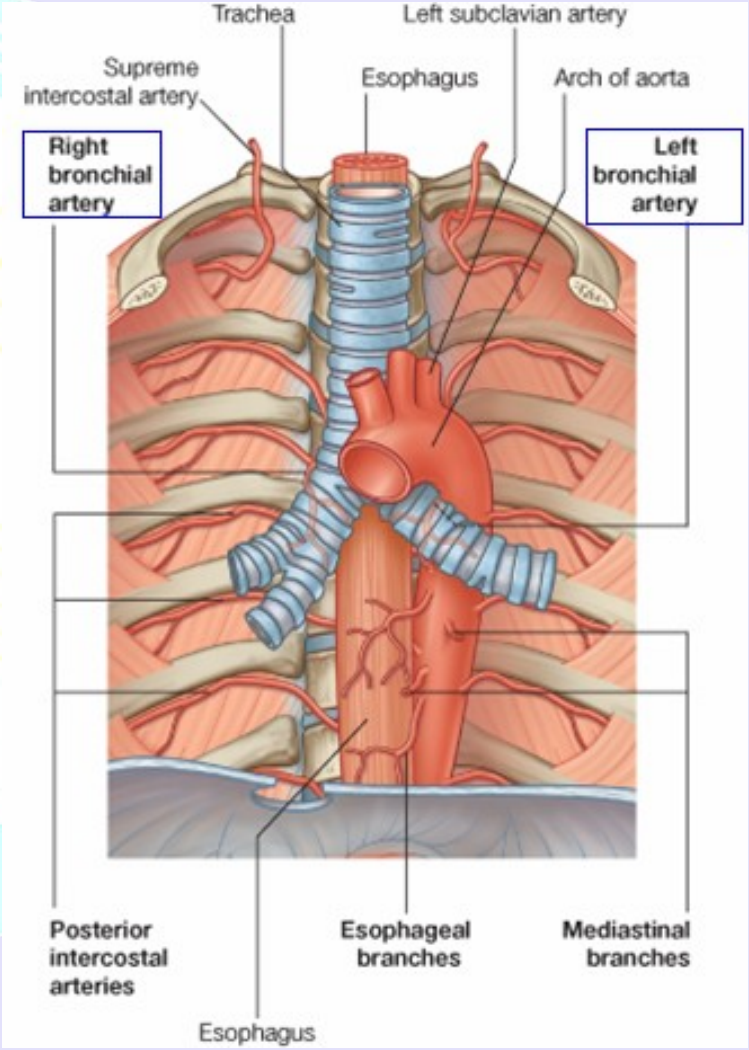
PULMONARY VEINS

- **Are intersegmental in drainage**
- **Leave lung as five pulmonary veins, one from each lobe of the lungs.**
- **However, the right upper and middle veins usually joins so that only four veins enter the left atrium**
- **Carry oxygenated blood from the respiratory part of the lung and deoxygenated part from the visceral pleura and from a part of bronchioles to the left atrium of the heart**



BRONCHIAL ARTERIES

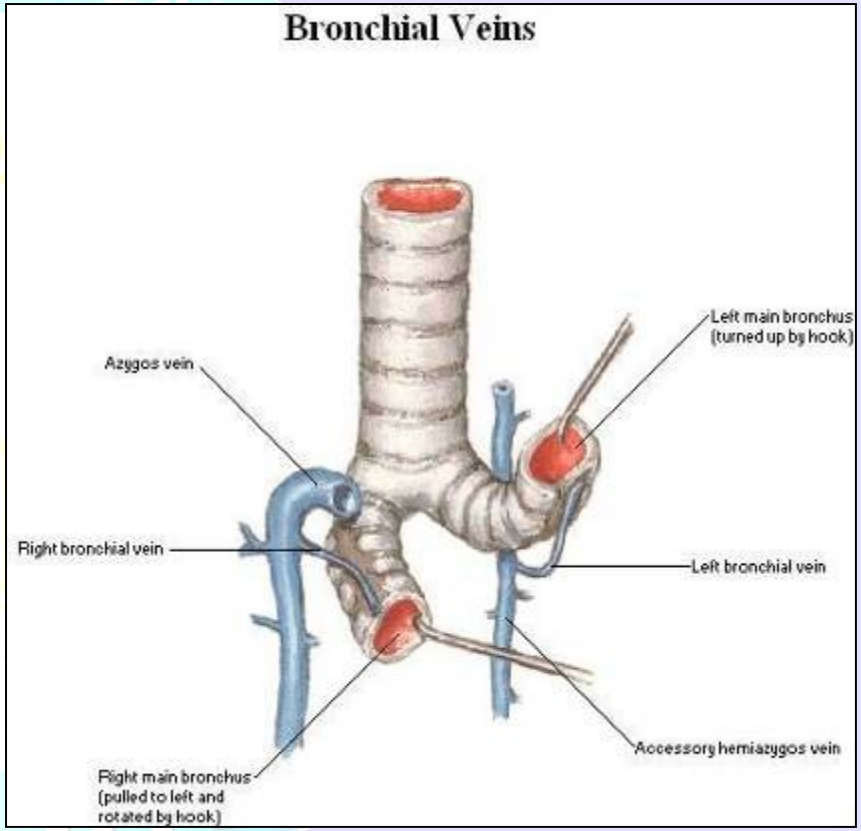
- Arise from thoracic aorta; usually one artery for right lung and two for the left lung
- Supply oxygenated blood to the non respiratory conducting tissues of the lungs and the visceral pleura
- Anastomosis occurs between capillaries of the bronchial and pulmonary systems



BRONCHIAL VEINS

- Receives blood from the bronchi

- Empty into the azygous vein on the right and accessory hemiazygous vein or superior intercostal vein on the left.



LYMPHATIC VESSELS

- Drain the bronchial tree, pulmonary vessels and connective tissue septa
- Run towards the hilus, drain to intapulmonary nodes →bronchopulmonary nodes → inferior and superior tracheobronchial nodes, paratracheal nodes, bronchomediastinal nodes and eventually to thoracic duct on the left and right lymphatic duct on the right
- Are not present in the walls of the pulmonary alveoli

NERVE SUPPLY

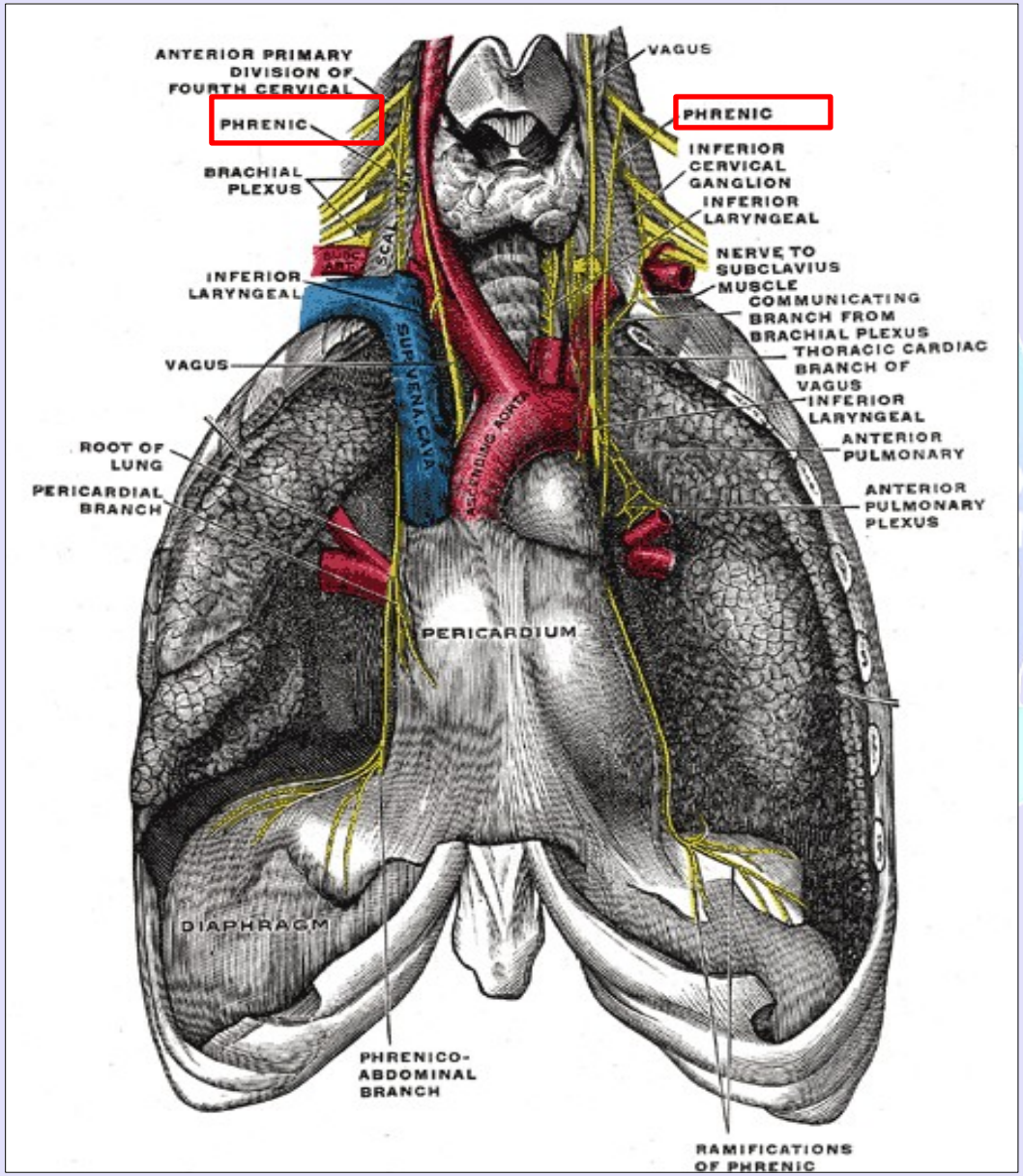
PULMONARY PLEXUS:

- Divided into anterior pulmonary plexus which lie in front of the root of lung
- Posterior pulmonary plexus ,which lie posterior to the root of the lung
- Has sympathetic fibers that dilate the lumina of the bronchi whereas parasympathetic fibres constrict the lumina and increase glandular secretions

PHRENIC NERVE

- Arises from C3-C5 cervical nerves and lies in front of anterior scalene muscle.
- Enters the thorax by passing deep to subclavian vein and superficial to subclavian arteries
- Runs anterior to the root of the lung ,whereas vagus nerve lies posterior to the root of the lung
- Innervates pericardium, the mediastinal and diaphragmatic pleura and the diaphragm

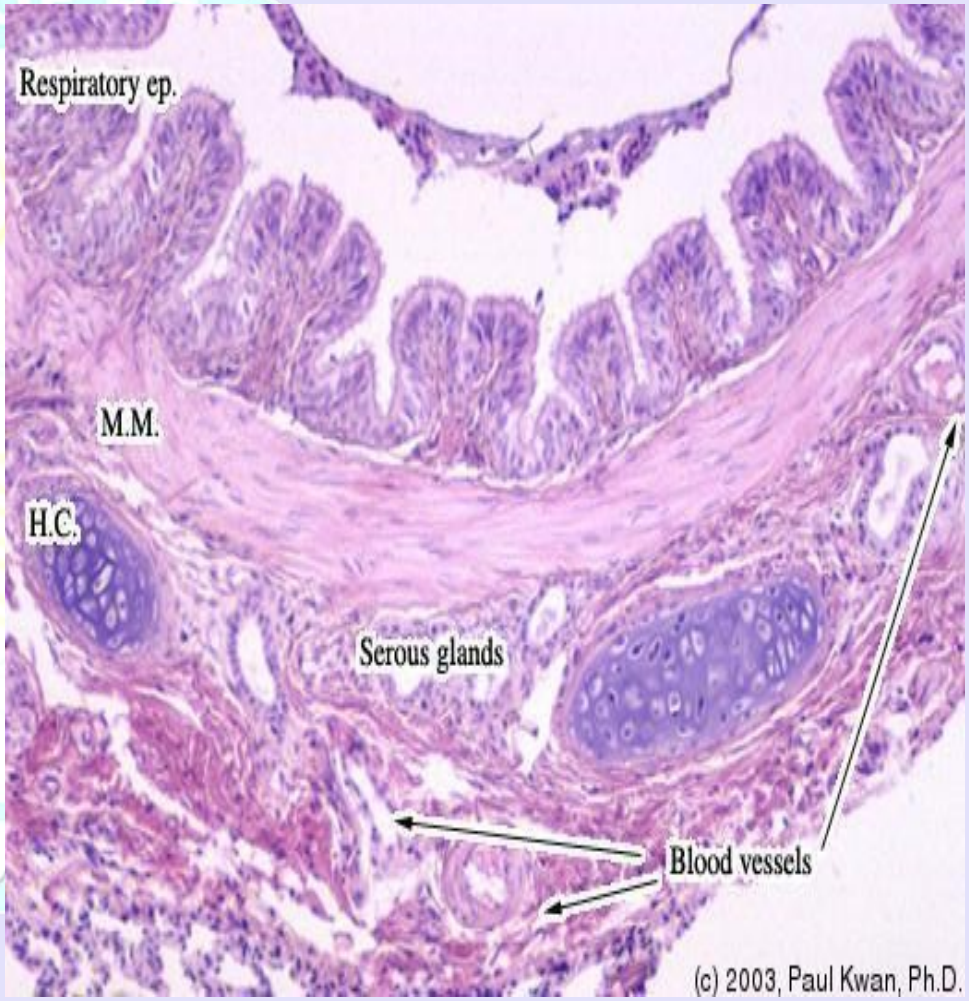
PHRENIC NERVE



HISTOLOGY OF LUNG

HISTOLOGY OF PRINCIPAL BRONCHI

- The principal bronchi has similar histological structure like trachea, but there difference between extra and intra pulmonary parts:
- Intra-pulmonary part has smaller diameter and more round as compare to the extra-pulmonary part of principal bronchus
- Contains smooth muscle in a layer
- Contains cartilages which are plate shaped

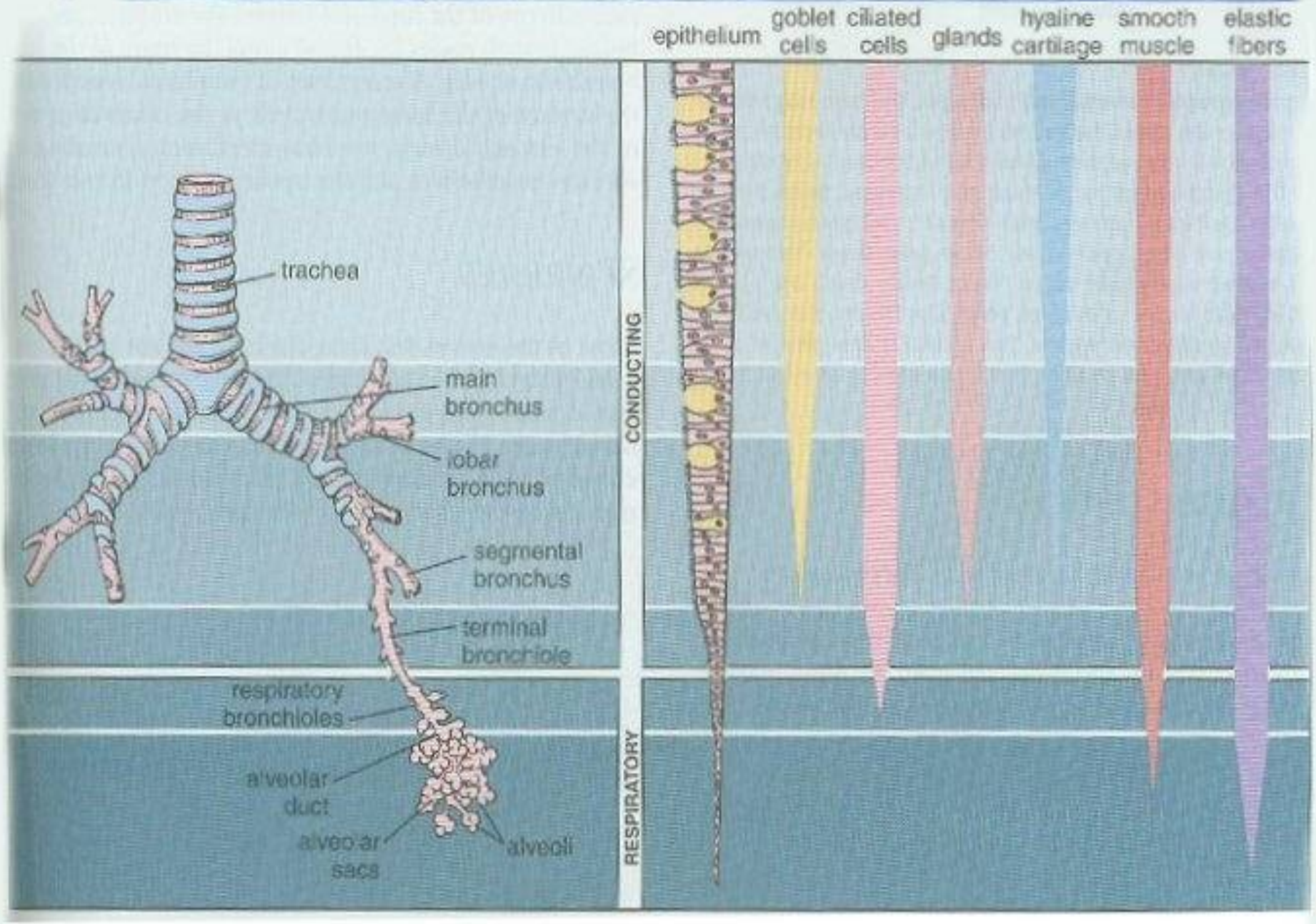


INTRA-PULMONARY BRONCHI

- The trachea bifurcates into two primary bronchi, which enter the lung and then branch several times to give rise to smaller secondary and tertiary bronchi [example] .
- Bronchi differ from the trachea in having plates rather than rings of cartilage, and in having a layer of smooth muscle between the lamina propria and submucosa.
- In smaller branches, the amount of cartilage decreases, whereas the amount of smooth muscle increases

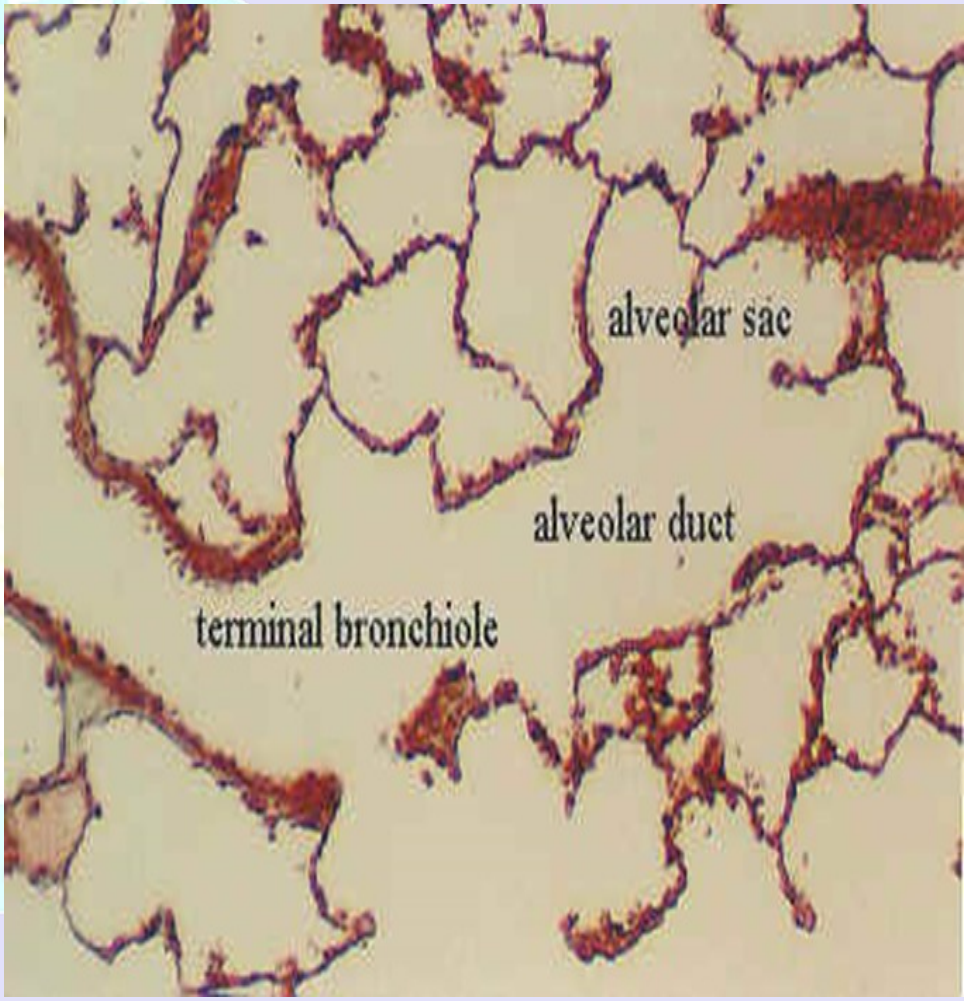


Divisions of the Bronchial Tree and Summary of Its Histologic Features



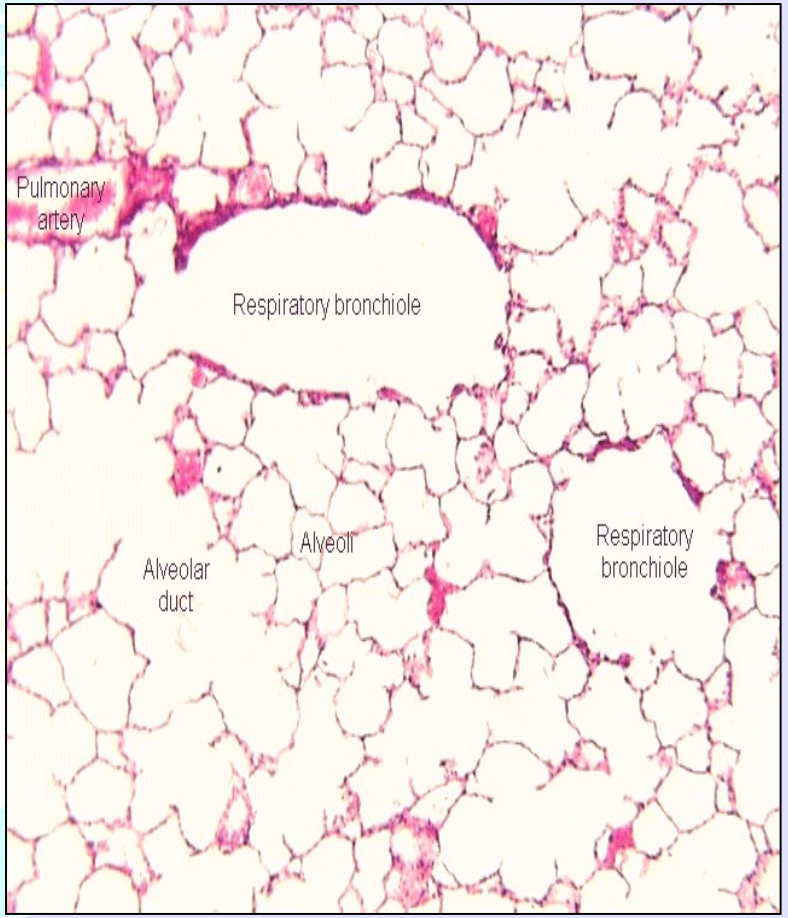
BRONCHIOLES

- Bronchioles are smaller branches of the bronchi, and are distinguished from them by the absence of cartilage and glands.
- In larger bronchioles, the epithelium is still ciliated, but is now usually simple columnar, whereas in the smallest bronchioles, the epithelium will be simple cuboidal (mostly Clara cells) and lack cilia altogether.
- The smooth muscle layer is generally quite prominent.
- The smallest conducting bronchioles consist of a simple cuboidal (or perhaps “low columnar”) epithelium of mostly Clara cells, a few ciliated cells, and NO goblet cells, and are called **Terminal Bronchioles**.



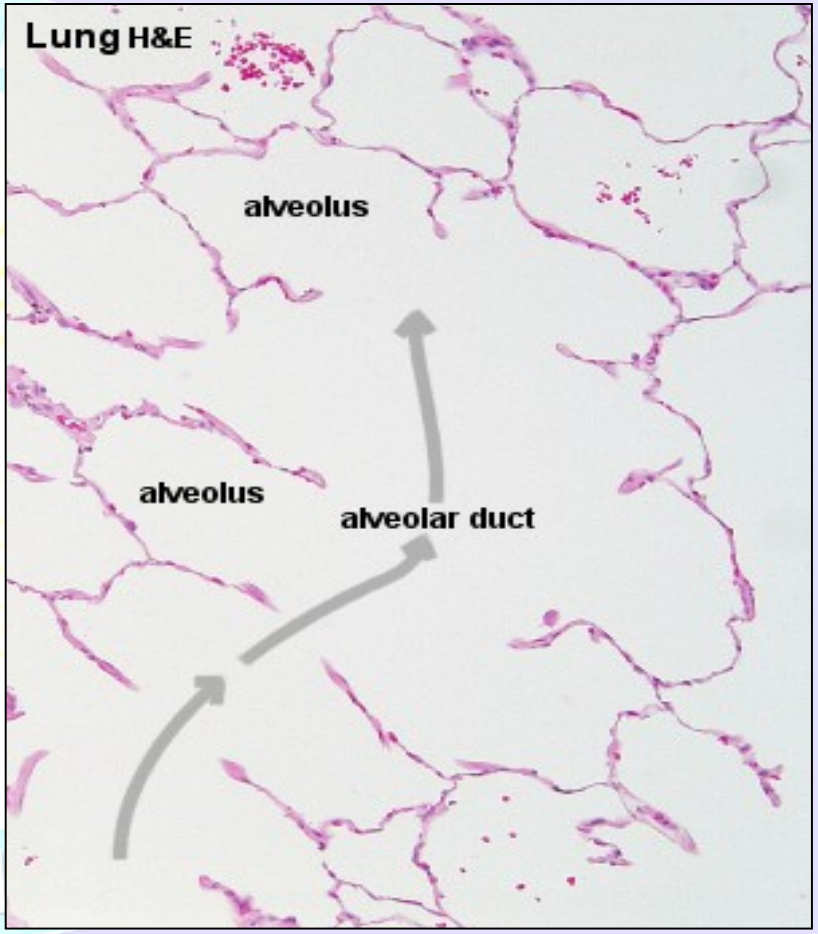
RESPIRATORY BRONCHIOLES

- Short, transitional regions of bronchioles which have alveoli in their walls.
- These bronchioles with alveoli in their walls are called respiratory bronchioles .
- They characteristically exhibit a progressive reduction in height of the epithelium and in the amount of smooth muscle between the openings of adjacent alveoli.



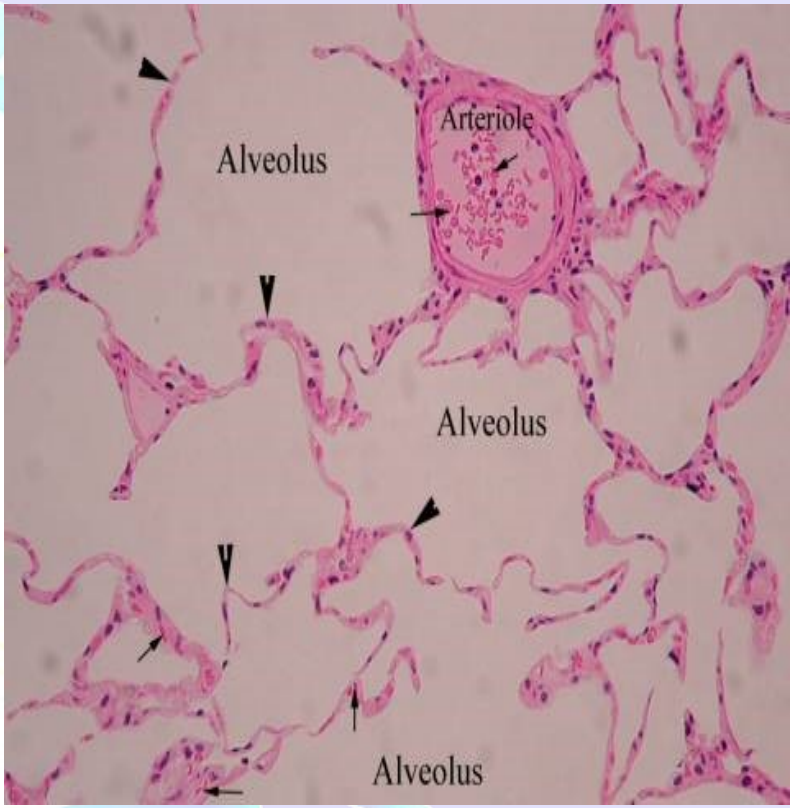
ALVEOLAR DUCTS

- The walls of alveolar ducts are so interrupted by alveoli and alveolar sacs (clusters of alveoli) that all that can be seen of the wall proper is small knobs of smooth muscle, collagen and elastic fibers.



ALVEOLUS

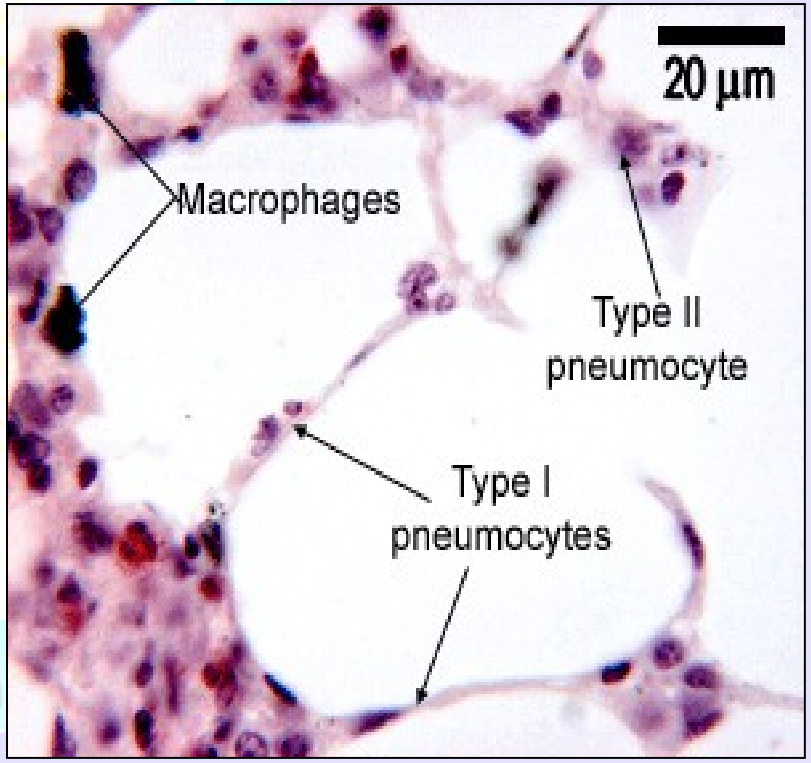
- The walls of these structures are covered on both sides by squamous epithelium
- Type I cells lining adjacent alveolar lumens. Within the walls is an extensive capillary network.
- The Type II pneumocytes which secrete surfactant, have large, rounded nuclei and vacuolated cytoplasm and are often difficult to identify in the light microscope (the “vacuoles” are actually granules of phospolipids that, unfortunately, are often extracted during tissue processing).



CELLS OF ALVEOLI

Type II pneumocytes with large round nuclei and clear cytoplasm are much more numerous and can be easily seen in the alveolar walls because they proliferate in an attempt to repair damaged alveoli.

Alveolar macrophages characteristic of this condition (also known as “heart failure” cells) become laden with brownish-black hemosiderin pigment resulting from the breakdown of erythrocytes leaking from the engorged capillaries.



INTERESTING FACTS

* At rest, the body takes in and breathes out about 10 liters of air each minute.

* The right lung is slightly larger than the left.

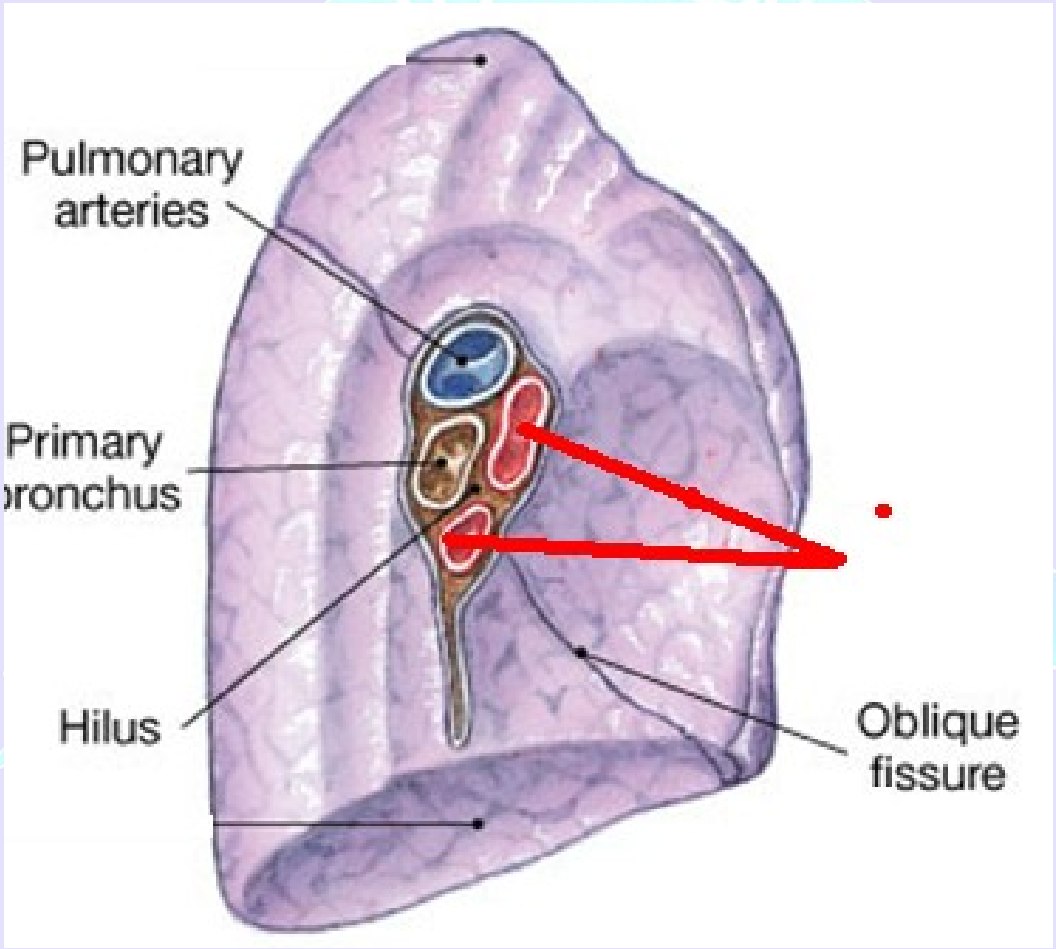
* The surface area of the lungs is roughly the same size as a tennis court.

* The capillaries in the lungs would extend 1,600 kilometers if placed end to end.

* A person at rest usually breathes between 12 and 15 times a minute.

* The breathing rate is faster in children and women than in men.

IDENTIFY THE LUNG(LEFT/RIGHT) NAME THE STRUCTURE MARKED WITH RED .



NAME THE FISSURES MARKED IN RED.

